

Course Plan

BCSE419L - Speech and Language Processing

(G1 + TG1) & (G2 + TG2) Slots

S.No.	Lecture Topic
1	Introductio Different Levels of
2	Complexities at each level of NLP
3	Morphology : Derivational and Inflectional Morphology
4	POS Tagging: Brill's tagger
5	Parsing: Difficulties with CFG, Chunking
6	Dependency Parsing
7	Semantics : Lexical and Compositional, WSD
8	Thematic Roles
9	Introduction to Corpora & Corpora Analysis
10	Sentence Segmentation
11	Stemming & Lemmatization
11	Bag of Words & Vector Space Model
12	N-gram Models
13	N-gram Models & perplexity, Smoothing
14	Word Embeddings: CBOW and Skip-Gram
15	Word Embeddings: Glove and Fasttext
16	Text Classification: Convolutional Neural Networks
17	Text Classification: Using Recurrent Neural Networks
18	Named Entity Recognition: CRF
19	Named Entity Recognition: LSTM
20	Encoder-decoder models & Attention in Deep learning
21	Introduction to Transformers and Other LLMs
22	Introduction to Transformers and Other LLMs
23	Introduction to Machine Translation: Simple and LLM methods
24	Text Summarization: Statistical and LLM methods

25	Question-Answering: Traditional Methods
26	Question-Answering: LLMs
27	Chatbot: with LLMs
28	Introduction to Speech Production : Place and Manner of Articulation
29	Phonetics and phonology
30	Vocal Tract Model
31	Characteristics of Speech Signal
32	Voice activity detection
33	STFT & MFCC
34	MFCC and Other feature representations
35	MFCC and Other feature representations
36	Features for Speaker & Music Recognition
37	Automatic Speech Recognition: Types and challenges
38	Isolated word Speech Recognition
39	LVCSR: HMM-GMM models for Speech Recognition
40	LVCSR: HMM-GMM models for Speech Recognition
41	HMM-DNN models for Speech Recognition
42	CTC models for Speech Recognition
43	Speaker Recognition
44	Guest Lecture
45	Guest Lecture

Course Assessment Procedure with Marks:

CAM: (60 Marks)	FAT (40 Marks)
CAT I - 15 Marks CAT II - 15 Marks Digital Assignment 1 : Individual Digital Assignment 2 : Project Submission (30 Marks) Digital Assignment 3:	

Note: Late submission of Assignments leads to loss of some % of Marks.